EXPERIMENT-30

30. CBC-MAC of a Two-Block Message

AIM

To write a C program demonstrating a weakness in basic CBC-MAC when the same key is used for messages of different lengths.

PROCEDURE

1. Select a secret key \(K\) and a one-block message \(X\).
2. Calculate the CBC-MAC of \(X\), denoted by \(T=E_K(X)\).
3. Construct a two-block message \(X\parallel(X\oplus T)\).
4. Calculate its CBC-MAC using the same key.
5. Observe that the final CBC-MAC is also \(T\), demonstrating a length-extension weakness in basic CBC-MAC.

CODE (C)

```
#include <stdio.h>

unsigned char encrypt_block(unsigned char block,
                            unsigned char key) {
    /* Toy block cipher for demonstration only */
    return (unsigned char)(block + key);
}

unsigned char cbc_mac(unsigned char *msg, int n,
                      unsigned char key) {
    unsigned char chain = 0;
    int i;

    for (i = 0; i < n; i++) {
        chain = encrypt_block(msg[i] ^ chain, key);
    }

    return chain;
}

int main(void) {
```

```
    unsigned char key = 0x2A;
    unsigned char X = 0x35;
    unsigned char T;
    unsigned char extended[2];
    unsigned char T2;

    /* MAC of the one-block message X */
    T = cbc_mac(&X, 1, key);

    /* Construct X || (X XOR T) */
    extended[0] = X;
    extended[1] = X ^ T;

    /* MAC of the two-block message */
    T2 = cbc_mac(extended, 2, key);

    printf("Key K = %02X\n", key);
    printf("Message X = %02X\n", X);
    printf("MAC of X, T = %02X\n", T);

    printf("Second block (X XOR T) = %02X\n", extended[1]);
    printf("Two-block message = %02X %02X\n",
           extended[0], extended[1]);
    printf("MAC of two-block message = %02X\n", T2);

    if (T == T2)
        printf("Both MAC values are equal.\n");

    return 0;
}
```

SAMPLE INPUT

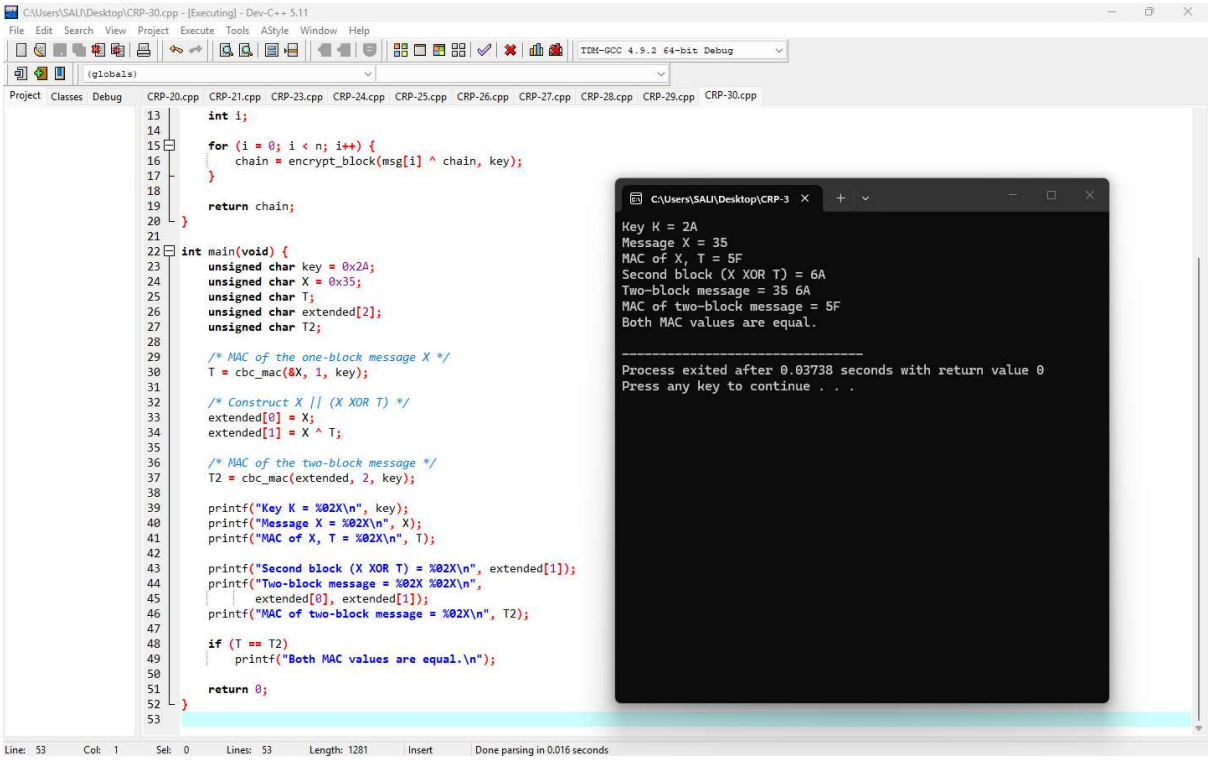
The values are initialized in the program.

```
Key K = 2A
Message X = 35
```

SAMPLE OUTPUT

```
Key K = 2A
Message X = 35
MAC of X, T = 5F
Second block (X XOR T) = 6A
Two-block message = 35 6A
```

```
MAC of two-block message = 5F
Both MAC values are equal.
```



RESULT

The program demonstrates that basic CBC-MAC is vulnerable when used directly for messages of different lengths. A secure implementation should use a properly designed construction such as CMAC or ensure message lengths are securely handled.